ML with Python (part1)

Installing Anaconda

Link to download Anaconda

https://www.anaconda.com/distribution/#windows

Steps to download Anaconda:

https://docs.anaconda.com/anaconda/install/windows/

Installing PyCharm Edu

https://www.jetbrains.com/pycharm-edu/

Exploring Anaconda and PyCharm

To be illustrated in the lecture.

Common Libraries

NumPy: It has advanced math functions, scientific computing package and useful features for operations on n-arrays and matrices in Python.

SciPy: SciPy is a library of software for engineering and science. SciPy contains modules for linear algebra, optimization, integration, and statistics. The main functionality of SciPy library is built upon NumPy, and thus, its arrays make substantial use of NumPy.

Pandas: It designed for quick and easy data manipulation, aggregation, and visualization.

Matplotlib: For visualization

Seaborn: For advanced visualization

SciKit-Learn:
Designed for specific functionalities like image processing and machine learning facilitation.

Theano, ***PyTorch,*** **TensorFlow**, and **Keras**: are used for Deep Learning**.**

NLTK and **Gensim**: are used for Natural Language Processing.

Selecting dataset

Iris.csv

Some basic operations over the dataset

Load The Data

We are using pandas to load the data. We will also use pandas next to explore the data both with descriptive statistics and data visualization.

```
import pandas

dataset = pandas.read_csv("C:\\Users\\USER\\iris_dataset.csv")
```

Dimensions of Dataset

```
print(dataset.shape)
```

Top rows of the Data

```
print(dataset.head(5))
```

Statistical Summary

```
print(dataset.describe( ))
```